

HW06 AI Audit Report

Declaration

I use AI tools for the following tasks: contract extraction and comparison; EP/BVA, state, security, and schema test ideation; generation and critique of draft test cases; Postman/Newman automation support; failure triage and bug-draft preparation; CI/CD configuration support; agent-generator architecture, pseudocode, reusable-skill authoring; and final-report drafting.

The evidenced AI tool is **OpenAI Codex**. The exact historical deployment/model version is not exposed by retained session metadata and is not invented. Postman, Newman 6.2.1, Node.js 20.20.2, GitHub Actions, Excel/openpyxl, PowerShell, and Git are execution/development tools rather than AI tools.

Source-log result

Source reviewed: [hw6/ai-audit-log.md](#).

The log indexes **38/38 planned phases (P0.1–P9.6)** and supplies each structural field, while preserving whether the underlying interaction was actually visible. An indexed reconstructed phase is not treated as an original chat record.

Provenance is not uniform:

- **15/38** records (P6.1–P9.6) are recoverable from visible session context.
- **23/38** records (P0.1–P5.6) use exact planned text and artifact mappings because their standalone retained chat messages are unavailable.
- **0/38** original per-message system timestamps are exposed.
- **0/38** exact historical model versions are exposed.

No timestamp schedule is reconstructed. All 38 timestamp fields are marked **UNAVAILABLE — ORIGINAL SYSTEM METADATA NOT EXPOSED**; file modification times are not substituted.

P0–P9 structural completeness

Phase	Planned	Indexed	Visible session	Playbook/artifact reconstructed	Evidence status
P0	3	3	0	3	RECONSTRUCTED
P1	4	4	0	4	RECONSTRUCTED
P2	4	4	0	4	RECONSTRUCTED
P3	4	4	0	4	RECONSTRUCTED
P4	2	2	0	2	RECONSTRUCTED
P5	6	6	0	6	RECONSTRUCTED
P6	2	2	2	0	PARTIAL METADATA
P7	4	4	4	0	PARTIAL METADATA
P8	3	3	3	0	PARTIAL METADATA

Phase	Planned	Indexed	Visible session	Playbook/artifact reconstructed	Evidence status
P9	6	6	6	0	PARTIAL METADATA
Total	38	38	15	23	INCOMPLETE ORIGINAL EVIDENCE

Short-message handling

Messages such as OKE, continue, status questions, and retry chatter were excluded because they produced no standalone artifact. Short decisions with material effect were retained inside the relevant interaction's human-review/correction field, including removal of PRD-C04, authorization to supplement missing cases, delivery of the real Postman screenshot, and acceptance of its visible-password limitation.

Compliance boundary

The audit file is complete only as a transparent **phase index**. It is not a fully compliant original interaction log: P0–P5 prompts are playbook/artifact reconstructions, and no original per-message timestamp or exact historical model version is available. A genuine platform export is required to close those gaps.

AI Critique

The AI was productive at generating breadth, but its output was not reliable enough to become a test oracle without human audit. The workbook contains 120 AI-generated cases, of which 89 were VALID, 7 INVALID, and 24 INCOMPLETE. These verdicts show a recurring weakness: the model often converted an unstated assumption into an exact boundary or expected status.

REG-AI-002 is a concrete example. The AI described a one-character name as the minimum boundary and expected HTTP 200, although the supplied contract did not define a minimum name length. The student marked the case INCOMPLETE, changed its technique from BVA to an equivalence partition for a valid non-empty string under the current contract, renamed the test accordingly, and retained the observable success and persistence checks. Similarly, REG-AI-003 assumed that an emoji must be accepted. The correction removed the emoji, used the clearly representable Unicode name “Nguyễn Anh,” and preserved UTF-8 round-trip verification.

The audit also refused to repair unsupported expectations silently. CPN-AI-016 remained INVALID because it demanded exact fractional monetary values while acknowledging that no rounding policy existed. It was not used as the oracle for the verified percentage-calculation defect; that defect instead used 500,000, whose 10% result is exact.

The main lesson is that AI is useful for proposing partitions, state paths, and security hypotheses, but not for deciding missing requirements. Each proposed oracle must be traced to a contract, ambiguous cases must remain visible, and execution evidence must be separated from test-quality verdicts. Human review should narrow scope, correct technique labels, and reject invented boundaries before automation or bug reporting.

Evidence: [23127334 HW06 AI Audit.md](#) and [23127334 HW06 API TestCases.xlsx](#).